Problems Section 1.3 / Charge & Current

1.1) $e = 1.6 \times 10^{-19} C$

a) $Q = N.e$ | N = Number of e^-
| $e = 1.6 \times 10^{-19}$

1.2) a) $I = \frac{dq}{dt}$

$I = \frac{d}{dt}(3t + 8)$ mC

$= 3$ mA

$= \boxed{3 \times 10^{-3} A}$

b) $I = \frac{d}{dt}(8t^2 + 4t - 2) C$

$= \boxed{16t + 4 \; C}$

c) $I = \frac{d}{dt}(3e^{-1} - 5e^{-2t})$ nC

$= -3e^{-1} + 10e^{-2t}$ nA

$= \boxed{(-3e^{-1} + 10e^{-2t}) \times 10^{-9} A}$

d) $I = \frac{d}{dt}(10 \sin 120\pi t)$ pC

$= 10 . \cos(120\pi t) \, 120\pi$ pA

$= \boxed{1200\pi \cos(120\pi 2) \times 10^{-12} A}$

e) $I = \frac{d}{dt}(20e^{-4t} \cos 50t) \mu C$

$= -80e^{-4t} (-50 \sin 50t) \mu A$

$= \boxed{(4000e^{-4t} \sin 50t) \times 10^{-6} A}$

1.3

a) $dq = i\,dt$

$\Rightarrow q = \int i\,dt$

$= \int 3\,dt$

$= 3t + c$

$\Rightarrow q(0) \Rightarrow 3(0) + c = 1$

$\Rightarrow c = 1$

$\therefore \boxed{q(t) = 3t + 1 \text{ A}}$

b) $q = \int i\,dt$

$= \int 2t + 5\,dt$

$= t^2 + 5t + c$

$q(0) \Rightarrow t^2 + 5t + c = 0$

$\Rightarrow c = 0$

$\boxed{\therefore q(t) = t^2 + 5t \text{ mA}}$

c) $q = \int i\,dt$

$= \int 20\cos(10t + \frac{\pi}{6})\,dt$

$= \dfrac{20\sin(10t + \frac{\pi}{6})}{10}$

$= 2\sin(10t + \frac{\pi}{6}) + c \ \mu C$

$\boxed{\int \cos(ax+b) = \frac{1}{a}\sin(ax+b)}$

$\therefore q(0) \Rightarrow 2\sin(10.0 + \frac{\pi}{6}) + c = 2$

$\Rightarrow 2 \cdot \frac{1}{2} + c = 2$

$\Rightarrow c = 2 - 1$

$= 1\,\mu C$

$\boxed{q(t) = 2\sin(10t + \frac{\pi}{6}) + 1 \ \mu A}$

d) $q = \int i \, dt$

$= \int 10e^{-30t} \sin 40t \, dt \quad \cancel{q(0)=0}$

$= 10 \cdot \dfrac{e^{-30t}\{-30\sin(40t) - 40\cos(40t)\}}{(-30)^2 + (40)^2} + c \quad \Bigg| \quad \int e^{at} \sin(bt)\,dt = \cancel{e^{at}} \dfrac{e^{at}\{a\sin(bt) - b\cos(bt)\}}{a^2+b^2}$

$q = \dfrac{10}{2500} e^{-30t}\{-30\sin(40t) - 40\cos 40t) + c$

$= e^{-30t}\{-0.12\sin(40t) - 0.16\cos(40t)\} + c$

$q(0) \Rightarrow e^{-30(0)}(-0.12\sin 0 - 0.16\cos 0) + c = 0$

$\Rightarrow -0.16 + c = 0$

$\Rightarrow c = 0.16$

$q(t) = e^{-30t}\{-0.12\sin(40t) - 0.16\cos(40t)\} + 0.16 \ \mu A$

$\boxed{1.4}$ $\Delta q = i \Delta t$

$q = 3.2 \times 20$

$= 64\ C$

$\boxed{1.5}$ $q = \int i\,dt$

$= \int_0^{10} \frac{1}{2} t\ dt$

$= \left[\frac{1}{4} t^2\right]_0^{10}$

$= \frac{1}{4}(10^2 - 0^2)$

$= 25\ C$

$\boxed{1.6}$ a) $I = \frac{dq}{dt} = \frac{\Delta q}{\Delta t} = \frac{q_2 - q_1}{t_2 - t_1}$

$= \frac{80 - 0\ mC}{2 - 0\ ms}$

$= 40\ A$

b) $\frac{80 - 80}{8 - 2} \frac{\cancel{\times 10^{-3}}}{\cancel{\times 10^{-3}}}$

$= 0\ A$

c) $\frac{0 - 80\ mC}{12 - 8\ ms}$

$= -20\ A$

1.7) $I_1 = \dfrac{q_2 - q_1}{t_2 - t_1}$ (ending, starting)

$= \dfrac{50-0}{2-0}$

$= 25A$

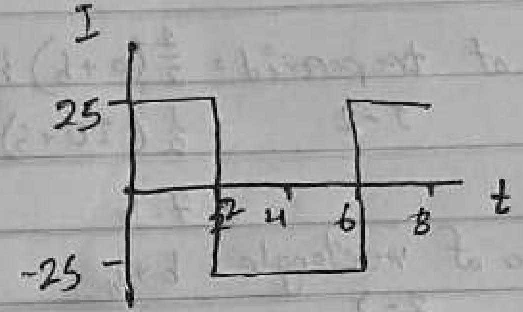
$I_2 = \dfrac{-50-50}{6-2}$

$= -25A$

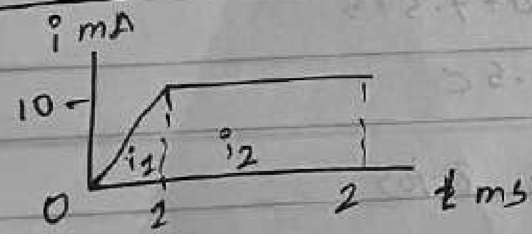
$I_3 = \dfrac{0-(-50)}{8-6}$

$= 25A$

graph



1.8



$\dfrac{x - x_1}{x_1 - x_2} = \dfrac{y - y_1}{y_1 - y_2}$ [total]

or

$\frac{1}{2} \times b \times h$ & $b \times h$

$\therefore q = \int i_1 dt + \int i_2 dt$

$= \int_0^1 10t\, dt + \int_1^2 10 dt$

$= \left[5t^2\right]_0^1 + \left[10t\right]_1^2$

$= 5 + [20-10]$

$= 15\ \mu C$

$q = i dt = mA \times ms$

$= \mu C$

$= 10^{-3} \times 10^{-3}$

$= 10^{-6}$

1.9 a) at $t = 1s$, from 0 to 1

$\therefore$ Area of rectangle $= b \times h$
0-1
$= 1 \times 10$
$= 10\ C$

b) at $t = 3s$, from 0 to 3

$\therefore$ Area of trapezoid $= \frac{1}{2}(a+b)h$
1-2
$= \frac{1}{2}(10+5)1$
$= 7.5$

$\therefore$ Area of rectangle $= b \times h$
2-3
$= 1 \times 5$
$= 5$

Total change $= 10 + 7.5 + 5$
$= 22.5\ C$

c) at $t = 5s$ from 0 to 5

$\therefore$ Area of rectangle $= b \times h$
3-4
$= 1 \times 5$
$= 5$

$\therefore$ Area of triangle $= \frac{1}{2} \times b \times h$
4-5
$= \frac{1}{2} \times 1 \times 3$
$= 2.5$

Total change $= 22.5 + 5 + 2.5$
$= 30\ C$

1.19 $P_{s_1} = -9 \times 4$

$= -36\,w$ (supply)

$P_{s_2} = 6 \times I$

$= 6I\,w$

$P_2 = 9\,w$

$P_1 = VI$

$= 3I\,w$

$\therefore 36 = 9 + 3I + 6I$

$\Rightarrow 27 = 9I$

$\Rightarrow I = 3\ A$

1.20 $P_{s_1} = -30 \times 6$

$= -180\ w$

$P_{s_2} = -5I_0 \times 3 \quad [I_0 = 2]$

$= -30\ w$

$\therefore P_1 = 12 \times 6$

$= 72\,w$

$P_2 = 3V_0 \quad P_3 = 56\ w \quad P_4 = 28\ w$

$\therefore 180 + 30 = 72 + 3V_0 + 56 + 28$

$\Rightarrow 210 = 156 + 3V_0$

$\Rightarrow V_0 = 18\ V$

Section 1.6 / Circuit Elements

1.16 $P = VI$

1.17 $P_1 + P_2 + P_3 + P_4 + P_5 = 0$

$\Rightarrow -205 + 60 + P_3 + 45 + 30 = 0$

$\Rightarrow P_3 = 70$ w (Received)

1.18 a)

$P_{s_1} = VI$

$= -9 \times 4$

$= -36$ w (supply)

$P_1 = VI$

$= 6 \times 4$

$= 24$ w (absorb)

$P_2 = VI$

$= 3 \times 4$

$= 12$ w (absorb)

$P_{s_1} = P_1 + P_2$

~~1.1~~ b)

$P_{s_1} = VI$

$= -24 \times 3$

$= -72$ w (supply)

$P_{s_2} = VI$

$= 3 I_0 \times 3$ $[I_0 = 3]$

$= 27$ w (absorb)

$P_1 = VI$

$= 10 \times 3$

$= 30$ w

$P_2 = VI$

$= 5 \times 3$

$= 15$ w

$P_{s_1} = P_1 + P_2 + P_{s_2}$

Comprehensive problems

1.32

$$I = \frac{\Delta q}{\Delta t}$$

$$\Rightarrow 20 \times 10^{-6} = \frac{15}{\Delta t}$$

$$\Rightarrow \Delta t = 750\,000 \text{ seconds}$$

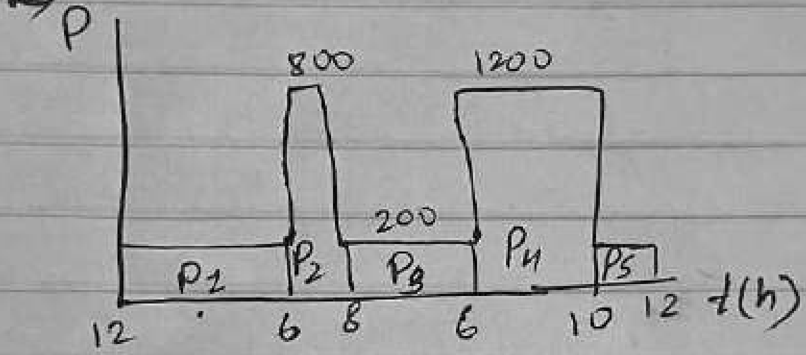
1.33

$$I = \frac{\Delta q}{\Delta t}$$

$$\Rightarrow 2 \times 10^{3} = \frac{\Delta q}{3 \times 10^{-3}}$$

$$\Rightarrow \Delta q = 6\,C$$

1.34 b)



$P_1 = 200 \times (12-6)$
$= 1200$ w

$P_2 = 800 \times (6-8)$
$= 1600$ w

$P_3 = 200 \times (8-6)$
$= 2000$ w

$P_4 = 1200 \times (6-10)$
$= 4800$ w

$P_5 = 200 \times (10-12)$
$= 400$ w

$$P_{total} = \frac{1200+1600+2000+4800+400}{24}$$

= Watt

$\boxed{1.36}$ a) $I = \frac{\Delta q}{\Delta t}$ $\Rightarrow I = \frac{160}{40}$

$\cancel{\Rightarrow \Delta q = 160 \times 40}$ $= 4\,A$

b) $I = \frac{\Delta q}{\Delta t}$

$\Rightarrow \Delta t = \frac{\Delta q}{I}$

$= \frac{160}{1 \times 10^{-3}}$

$= 160000$ hours

$\boxed{1.39}$

(a) True (b) False

Answers: 1.1b, 1.2a, 1.3c, 1.4a, 1.5b, 1.6c, 1.7a, 1.8c, 1.9b, 1.10d.

Problems

Section 1.3 Charge and Current

1.1 How many coulombs are represented by these amounts of electrons?

(a) 6.482×10^{17} (b) 1.24×10^{18}

(c) 2.46×10^{19} (d) 1.628×10^{20}

1.2 Determine the current flowing through an element if the charge flow is given by

(a) $q(t) = (3t + 8)$ mC

(b) $q(t) = (8t^2 + 4t - 2)$ C

(c) $q(t) = (3e^{-t} - 5e^{-2t})$ nC

(d) $q(t) = 10 \sin 120\pi t$ pC

(e) $q(t) = 20e^{-4t} \cos 50t\ \mu$C

1.3 Find the charge $q(t)$ flowing through a device if the current is:

(a) $i(t) = 3$ A, $q(0) = 1$ C

(b) $i(t) = (2t + 5)$ mA, $q(0) = 0$

(c) $i(t) = 20\cos(10t + \pi/6)\ \mu$A, $q(0) = 2\ \mu$C

(d) $i(t) = 10e^{-30t} \sin 40t$ A, $q(0) = 0$

1.4 A current of 3.2 A flows through a conductor. Calculate how much charge passes through any cross-section of the conductor in 20 s.

1.5 Determine the total charge transferred over the time interval of $0 \le t \le 10$ s when $i(t) = \frac{1}{2}t$ A.

1.6 The charge entering a certain element is shown in Fig. 1.23. Find the current at:

(a) $t = 1$ ms (b) $t = 6$ ms (c) $t = 10$ ms

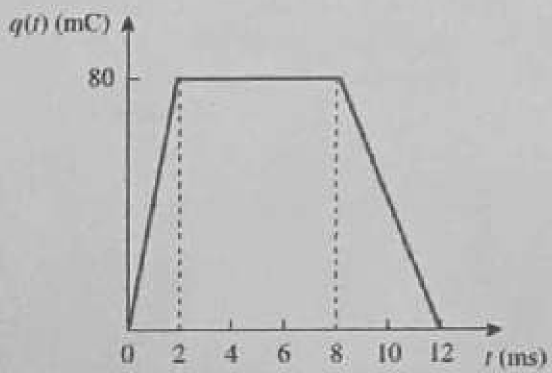
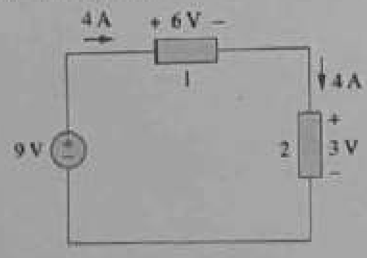


Figure 1.23
For Prob. 1.6.

1.18 Calculate the power absorbed or supplied by each element in Fig. 1.29.



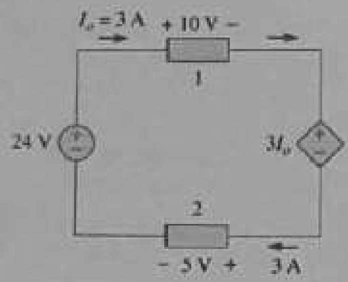


Figure 1.29
For Prob. 1.18.

1.19 Find I in the network of Fig. 1.30.

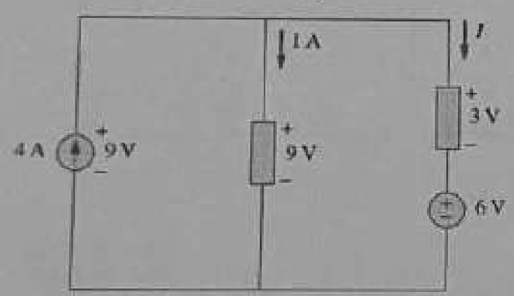


Figure 1.30
For Prob. 1.19.

1.20 Find V_o in the circuit of Fig. 1.31.

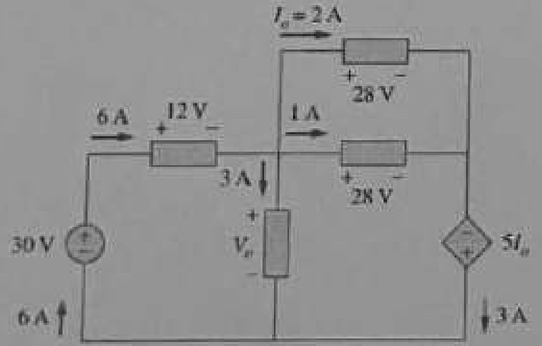


Figure 1.31
For Prob. 1.20.

Section 1.7 Applications

1.21 A 60-W incandescent bulb operates at 120 V. How many electrons and coulombs flow through the bulb in one day?

1.22 A lightning bolt strikes an airplane with 30 kA for 2 ms. How many coulombs of charge are deposited on the plane?

1.23 A 1.8-kW electric heater takes 15 min to boil a quantity of water. If this is done once a day and power costs 10 cents/kWh, what is the cost of its operation for 30 days?

1.24 A utility company charges 8.5 cents/kWh. If a consumer operates a 40-W light bulb continuously for one day, how much is the consumer charged?

1.25 A 1.2-kW toaster takes roughly 4 minutes to heat four slices of bread. Find the cost of operating the toaster once per day for 1 month (30 days). Assume energy costs 9 cents/kWh.

1.26 A 12-V car battery supported a current of 150 mA to a bulb. Calculate:

(a) the power absorbed by the bulb.

(b) the energy absorbed by the bulb over an interval of 20 minutes.

1.27 A constant current of 3 A for 4 hours is required to charge an automotive battery. If the terminal voltage is $10 + t/2$ V, where t is in hours,

(a) how much charge is transported as a result of the charging?

(b) how much energy is expended?

(c) how much does the charging cost? Assume electricity costs 9 cents/kWh.

1.28 A 30-W incandescent lamp is connected to a 120-V source and is left burning continuously in an otherwise dark staircase. Determine:

(a) the current through the lamp.

(b) the cost of operating the light for one non-leap year if electricity costs 12 cents per kWh.

1.29 An electric stove with four burners and an oven is used in preparing a meal as follows.

Burner 1: 20 minutes Burner 2: 40 minutes
Burner 3: 15 minutes Burner 4: 45 minutes
Oven: 30 minutes

If each burner is rated at 1.2 kW and the oven at 1.8 kW, and electricity costs 12 cents per kWh, calculate the cost of electricity used in preparing the meal.

1.7 The charge flowing in a wire is plotted in Fig. 1.24. Sketch the corresponding current.

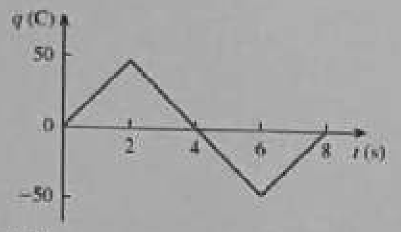


Figure 1.24
For Prob. 1.7.

1.8 The current flowing past a point in a device is shown in Fig. 1.25. Calculate the total charge through the point.

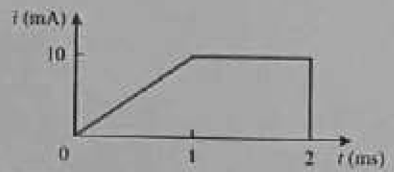


Figure 1.25
For Prob. 1.8.

1.9 The current through an element is shown in Fig. 1.26. Determine the total charge that passed through the element at:

(a) $t = 1$ s (b) $t = 3$ s (c) $t = 5$ s

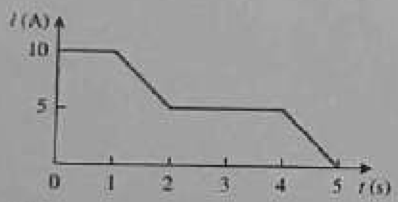


Figure 1.26
For Prob. 1.9.

Sections 1.4 and 1.5 Voltage, Power, and Energy

1.10 A lightning bolt with 8 kA strikes an object for 15 μs. How much charge is deposited on the object?

1.11 A rechargeable flashlight battery is capable of delivering 85 mA for about 12 h. How much charge can it release at that rate? If its terminal voltage is 1.2 V, how much energy can the battery deliver?

1.12 If the current flowing through an element is given by

$$i(t) = \begin{cases} 3t\text{A}, & 0 \leqq t < 6 \text{ s} \\ 18\text{A}, & 6 \leqq t < 10 \text{ s} \\ -12\text{A}, & 10 \leqq t < 15 \text{ s} \\ 0, & t \geqq 15 \text{ s} \end{cases}$$

Plot the charge stored in the element over $0 < t < 20$ s.

1.13 The charge entering the positive terminal of an element is

$$q = 10 \sin 4\pi t \text{ mC}$$

while the voltage across the element (plus to minus) is

$$v = 2 \cos 4\pi t \text{ V}$$

(a) Find the power delivered to the element at $t = 0.3$ s.

(b) Calculate the energy delivered to the element between 0 and 0.6 s.

1.14 The voltage v across a device and the current i through it are

$$v(t) = 5 \cos 2t \text{ V}, \qquad i(t) = 10(1 - e^{-0.5t}) \text{ A}$$

Calculate:

(a) the total charge in the device at $t = 1$ s

(b) the power consumed by the device at $t = 1$ s.

1.15 The current entering the positive terminal of a device is $i(t) = 3e^{-2t}$ A and the voltage across the device is $v(t) = 5di/dt$ V.

(a) Find the charge delivered to the device between $t = 0$ and $t = 2$ s.

(b) Calculate the power absorbed.

(c) Determine the energy absorbed in 3 s.

Section 1.6 Circuit Elements

1.16 Find the power absorbed by each element in Fig. 1.27.

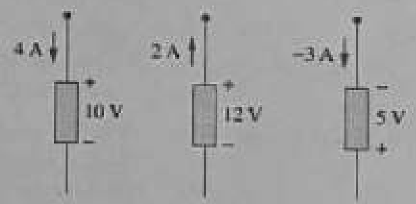


Figure 1.27
For Prob. 1.16.

1.17 Figure 1.28 shows a circuit with five elements. If $p_1 = -205$ W, $p_2 = 60$ W, $p_4 = 45$ W, $p_5 = 30$ W, calculate the power p_3 received or delivered by element 3.

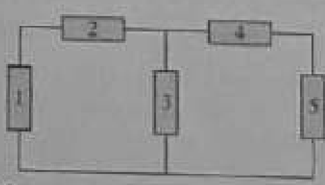


Figure 1.28
For Prob. 1.17.

1.30 Reliant Energy (the electric company in Houston, Texas) charges customers as follows:

Monthly charge \$6

First 250 kWh @ \$0.02/kWh

All additional kWh @ \$0.07/kWh

If a customer uses 1,218 kWh in one month, how much will Reliant Energy charge?

1.31 In a household, a 120-W personal computer (PC) is run for 4 h/day, while a 60-W bulb runs for 8 h/day. If the utility company charges \$0.12/kWh, calculate how much the household pays per year on the PC and the bulb.

Comprehensive Problems

1.32 A telephone wire has a current of 20 μA flowing through it. How long does it take for a charge of 15 C to pass through the wire?

1.33 A lightning bolt carried a current of 2 kA and lasted for 3 ms. How many coulombs of charge were contained in the lightning bolt?

1.34 Figure 1.32 shows the power consumption of a certain household in 1 day. Calculate:

(a) the total energy consumed in kWh.

(b) the average power per hour.

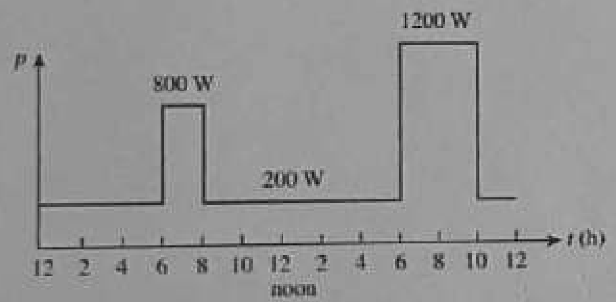


Figure 1.32
For Prob. 1.34.

1.35 The graph in Fig. 1.33 represents the power drawn by an industrial plant between 8:00 and 8:30 A.M. Calculate the total energy in MWh consumed by the plant.

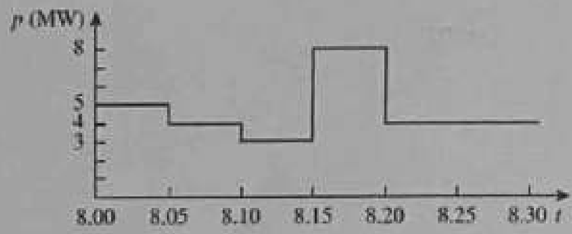


Figure 1.33
For Prob. 1.35.

1.36 A battery may be rated in ampere-hours (Ah). A lead-acid battery is rated at 160 Ah.

(a) What is the maximum current it can supply for 40 h?

(b) How many days will it last if it is discharged at 1 mA?

1.37 A unit of power often used for electric motors is the horsepower (hp), which equals 746 W. A small electric car is equipped with a 40-hp electric motor. How much energy does the motor deliver in one hour, assuming the motor is operating at maximum power for the whole time?

1.38 How much energy does a 10-hp motor deliver in 30 minutes? Assume that 1 horsepower = 746 W.

1.39 A 600-W TV receiver is turned on for 4 h with nobody watching it. If electricity costs 10 cents/kWh, how much money is wasted?